Staggered mesh method for GW calculation

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1 Theory

1.1 Self-energy in imaginary frequency domain

The GW self-energy entry in imaginary frequency domain can be written using planewave basis as

$$\begin{aligned}\boldsymbol{\Sigma}_{nn'}(\mathbf{k},\mathrm{i}\omega) &= -\frac{1}{2\pi N_{\mathbf{k}}\Omega}\sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{0}}\sum_{m\mathbf{G}\mathbf{G}'}\int_{-\infty}^{\infty}\mathrm{d}\omega' W_{\mathbf{G}\mathbf{G}'}\left(\mathbf{q},\mathrm{i}\omega'\right)\frac{\rho_{m\mathbf{k}-\mathbf{q}}^{n\mathbf{k}}(\mathbf{G})\rho_{m\mathbf{k}-\mathbf{q}}^{n'\mathbf{k}^{*}}\left(\mathbf{G}'\right)}{\mathrm{i}\left(\omega+\omega'\right)+\epsilon_{\mathrm{F}}-\epsilon_{m\mathbf{k}-\mathbf{q}}} \\ &= -\frac{1}{(2\pi)^{4}}\int_{-\infty}^{\infty}\mathrm{d}\omega'\frac{|\Omega^{*}|}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{0}}\left(\sum_{m\mathbf{G}\mathbf{G}'}W_{\mathbf{G}\mathbf{G}'}\left(\mathbf{q},\mathrm{i}\omega'\right)\frac{\rho_{m\mathbf{k}-\mathbf{q}}^{n\mathbf{k}}(\mathbf{G})\rho_{m\mathbf{k}-\mathbf{q}}^{n'\mathbf{k}^{*}}\left(\mathbf{G}'\right)}{\mathrm{i}\left(\omega+\omega'\right)+\epsilon_{\mathrm{F}}-\epsilon_{m\mathbf{k}-\mathbf{q}}}\right). \end{aligned} \tag{1}$$

The symmetric screened Coulomb operator is defined as

$$\begin{aligned} W_{\mathbf{G}\mathbf{G}'}(\mathbf{q},\mathrm{i}\omega') &= \frac{\sqrt{4\pi}}{|\mathbf{q}+\mathbf{G}|}\epsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q},\mathrm{i}\omega')\frac{\sqrt{4\pi}}{|\mathbf{q}+\mathbf{G}'|}-\delta_{\mathbf{G}\mathbf{G}'}\frac{4\pi}{|\mathbf{q}+\mathbf{G}|^{2}}, \\ \epsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q},\mathrm{i}\omega') &= \delta_{\mathbf{G}\mathbf{G}'}-\frac{\sqrt{4\pi}}{|\mathbf{q}+\mathbf{G}|}\chi_{\mathbf{G}\mathbf{G}'}(\mathbf{q},\mathrm{i}\omega')\frac{\sqrt{4\pi}}{|\mathbf{q}+\mathbf{G}'|}, \\ \chi_{\mathbf{G}\mathbf{G}'}(\mathbf{q},\mathrm{i}\omega') &= \frac{4}{N_{\mathbf{k}}\Omega}\sum_{\mathbf{k}\in\mathcal{K}}\sum_{i}^{\mathrm{occ}}\sum_{a}^{\mathrm{vir}}\rho_{a\mathbf{k}-\mathbf{q}}^{i\mathbf{k}}(\mathbf{G})\frac{\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}-\mathbf{q}}}{\omega'^{2}+\left(\epsilon_{i\mathrm{k}}-\epsilon_{a\mathbf{k}-\mathbf{q}}\right)^{2}}\rho_{a\mathbf{k}-\mathbf{q}}^{i\mathbf{k}*}\left(\mathbf{G}'\right), \\ \rho_{m(\mathbf{k}-\mathbf{q})}^{n\mathbf{k}}(\mathbf{G}) &= \langle\psi_{n\mathbf{k}}|e^{\mathrm{i}(\mathbf{q}+\mathbf{G})\mathbf{r}}|\psi_{m(\mathbf{k}-\mathbf{q})}\rangle=\int_{\Omega}u_{n\mathbf{k}}^{*}(\mathbf{r})u_{m(\mathbf{k}-\mathbf{q})}(\mathbf{r})e^{\mathrm{i}\mathbf{G}\mathbf{r}}\,\mathrm{d}\mathbf{r}. \end{aligned}$$

In standard GW, the self-energy entries $\boldsymbol{\Sigma}_{nn'}(\mathbf{k},\mathrm{i}\omega)$ have $\mathbf{k}$ sampled on an arbitrary MP mesh $\mathcal{K}$ in Ω^{*}. The same mesh $\mathcal{K}$ is also used to sample all momentum vectors $\mathbf{k}-\mathbf{q}$ in the calculation. As a result, the induced mesh $\mathcal{K}_{\mathbf{q}}^{0}$ in Eq. (1) for momentum transfer $\mathbf{q}$ is of the same shape as $\mathcal{K}$ and always contains the Γ point.

1.2 Thermodynamic limit and finite size error

For each $\boldsymbol{\Sigma}_{nn'}(\mathbf{k},\mathrm{i}\omega)$, we hide all the dependence on $\mathbf{k},\omega,n,n'$ for brevity and denote the bracketed summation in Eq. (1) as

$$M_{N_{\mathbf{k}}}(\omega')=\frac{|\Omega^{*}|}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{0}}\left(\sum_{m\mathbf{G}\mathbf{G}'}W_{\mathbf{G}\mathbf{G}'}\left(\mathbf{q},\mathrm{i}\omega'\right)\frac{\rho_{m\mathbf{k}-\mathbf{q}}^{n\mathbf{k}}(\mathbf{G})\rho_{m\mathbf{k}-\mathbf{q}}^{n'\mathbf{k}^{*}}\left(\mathbf{G}'\right)}{\mathrm{i}\left(\omega+\omega'\right)+\epsilon_{\mathrm{F}}-\epsilon_{m\mathbf{k}-\mathbf{q}}}\right):=\frac{|\Omega^{*}|}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{0}}f(\mathbf{q};\omega'). \tag{2}$$

In the TDL, this intermediate quantity converges to

$$M_{\mathrm{TDL}}(\omega') = \lim_{N_{\mathbf{k}}\to\infty} M_{N_{\mathbf{k}}}(\omega') = \int_{\Omega^*} \mathrm{d}\mathbf{q} f(\mathbf{q};\omega'). \tag{3}$$

The self energy entry in the TDL can then be formulated as

$$\mathbf{\Sigma}^{\mathrm{TDL}}_{nn'}(\mathbf{k},\mathrm{i}\omega) = -\frac{1}{(2\pi)^4}\int_{-\infty}^{\infty} \mathrm{d}\omega' M_{\mathrm{TDL}}(\omega'). \tag{4}$$

Note that $\chi_{\mathbf{GG}'}, \epsilon_{\mathbf{GG}'}, W_{\mathbf{GG}}$, and f all depend on the $\mathbf{k}$-point mesh $\mathcal{K}$, and these operators and functions converge to exact values in the TDL. Specifically, the linear response operator in the TDL converges to

$$\chi^{\mathrm{TDL}}_{\mathbf{GG}'}(\mathbf{q},\mathrm{i}\omega') = \frac{2}{(2\pi)^3}\int_{\Omega^*} \mathrm{d}\mathbf{k} \sum_{i}^{\mathrm{occ}}\sum_{a}^{\mathrm{vir}} \rho^{i\mathbf{k}}_{a\mathbf{k}-\mathbf{q}}(\mathbf{G}) \frac{\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}-\mathbf{q}}}{\omega'^2 + \left(\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}-\mathbf{q}}\right)^2} \rho^{i\mathbf{k}*}_{a\mathbf{k}-\mathbf{q}}(\mathbf{G}'). \tag{5}$$

Then $\epsilon_{\mathbf{GG}'}, W_{\mathbf{GG}}$, and f in the TDL can be defined accordingly using $\chi^{\mathrm{TDL}}_{\mathbf{GG}'}(\mathbf{q},\mathrm{i}\omega')$.

Overall, one major source of the finite size error in self-energy energy calculation, i.e., $\mathbf{\Sigma}^{\mathrm{TDL}}_{nn'}(\mathbf{k},\mathrm{i}\omega) - \mathbf{\Sigma}_{nn'}(\mathbf{k},\mathrm{i}\omega)$, is the composition of all the quadrature errors in estimating the three integrals above using trapezoidal rules:

1. Integrating over $\mathbf{q}$ to compute $M_{\mathrm{TDL}}(\omega')$ in Eq. (3) for each ω';
2. Integrating over ω' to compute $\mathbf{\Sigma}^{\mathrm{TDL}}_{nn'}(\mathbf{k},\mathrm{i}\omega)$ in Eq. (4).
3. Integrating over $\mathbf{k}$ to compute $\chi^{\mathrm{TDL}}_{\mathbf{GG}'}(\mathbf{q},\mathrm{i}\omega')$ in Eq. (5) for each $\mathbf{q}$ and ω';

Assuming that the quadrature errors from the last two approximations above are negligible, we focus on the numerical quadrature for estimating $M_{\mathrm{TDL}}(\omega')$ in Eq. (3) with integrand

$$f(\mathbf{q};\omega') = \sum_{m\mathbf{GG}'} W_{\mathbf{GG}'}(\mathbf{q},\mathrm{i}\omega') \frac{\rho^{n\mathbf{k}}_{m\mathbf{k}-\mathbf{q}}(\mathbf{G})\rho^{n'\mathbf{k}^*}_{m\mathbf{k}-\mathbf{q}}(\mathbf{G}')}{\mathrm{i}(\omega+\omega') + \epsilon_{\mathrm{F}} - \epsilon_{m\mathbf{k}-\mathbf{q}}}. \tag{6}$$

2 Staggered mesh method

The staggered mesh method below is proposed to reduce the finite size error in computing $M_{\mathrm{TDL}}(\omega')$. Using the analysis in Appendix .1, $f(\mathbf{q};\omega')$ can be shown to be periodic with respect to $\mathbf{q}\in\Omega^*$ and smooth everywhere except at $\mathbf{q}=\mathbf{0}$ where it has asymptotic nonsmooth terms as

$$f(\mathbf{q};\omega') = \delta_{nn'}\frac{4\pi}{\mathrm{i}(\omega+\omega') + \epsilon_{\mathrm{F}} - \epsilon_{n\mathbf{k}}}\left(\frac{1}{\mathbf{q}^T\mathbf{L}_{\omega'}\mathbf{q}} - \frac{1}{|\mathbf{q}|^2}\right) + \frac{\mathcal{O}(|\mathbf{q}|^2)}{|\mathbf{q}|^2} + \frac{\mathcal{O}(|\mathbf{q}|^2)}{\mathbf{q}^T\mathbf{L}_{\omega'}\mathbf{q}} + \text{smooth terms}.$$

Our recent error analysis result [2] suggests that these nonsmooth terms of f determines the asymptotic magnitude of the quadrature error.

The staggered mesh method consists of two components: application of a singularity subtraction method to remove the dominant $\mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}})$ quadrature error due to leading nonsmooth terms $\frac{\mathcal{O}(1)}{|\mathbf{q}|^2}$ and $\frac{\mathcal{O}(1)}{\mathbf{q}^T L_{\omega'}\mathbf{q}}$, and application of a non-Γ-centered mesh $\mathcal{K}^{\frac{1}{2}}_{\mathbf{q}}$ for the trapezoidal rule to remove a

significant portion of the $\mathcal{O}(N_{\mathbf{k}}^{-1})$ quadrature error due to second leading nonsmooth terms $\frac{\mathcal{O}(|\mathbf{q}|^2)}{|\mathbf{q}|^2}$ and $\frac{\mathcal{O}(|\mathbf{q}|^2)}{\mathbf{q}^T L_{\omega'}\mathbf{q}}$.

Different from standard calculation in Eq. (2), we estimate $M_{\mathrm{TDL}}(\omega')$ as

$$\begin{aligned} M_{\mathrm{TDL}}(\omega') &\approx \frac{|\Omega^*|}{N_{\mathbf{k}}} \sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}} f(\mathbf{q};\omega') + \delta_{nn'}\left(\int_{\Omega^*} \mathrm{d}\mathbf{q} - \frac{|\Omega^*|}{N_{\mathbf{k}}} \sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}}\right) f_{\mathrm{aux}}(\mathbf{q};\omega') \\ &:= M_{N_{\mathbf{k}}}^{\mathrm{stagger}}(\omega') + \delta_{nn'}\xi_{N_{\mathbf{k}}}^{\mathrm{stagger}}(\omega') \end{aligned}$$

where $\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}$ is a $\mathcal{K}$-shaped mesh from the half-mesh-size shift of a Γ-centered mesh. Auxiliary function $f_{\mathrm{aux}}(\mathbf{q})$ is periodic and shares the same leading nonsmooth terms as $f(\mathbf{q};\omega')$, and is defined as

$$f_{\mathrm{aux}}(\mathbf{q};\omega') = \frac{4\pi}{\mathrm{i}\,(\omega+\omega') + \epsilon_{\mathrm{F}} - \epsilon_{n\mathbf{k}}}\left(\sum_{\mathbf{G}\in\mathbb{L}^*} \frac{1}{(\mathbf{q}+\mathbf{G})^T\mathbf{L}_{\omega'}(\mathbf{q}+\mathbf{G})} - \sum_{\mathbf{G}\in\mathbb{L}^*}\frac{1}{|\mathbf{q}+\mathbf{G}|^2}\right). \tag{7}$$

Plugging this quadrature scheme into self-energy calculation, we get the staggered mesh method as

$$\begin{aligned} \mathbf{\Sigma}_{nn'}^{\mathrm{stagger}}(\mathbf{k},\mathrm{i}\omega) = -\frac{1}{(2\pi)^4}\int_{-\infty}^{\infty}\mathrm{d}\omega'\Bigg[&\frac{|\Omega^*|}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}}\left(\sum_{m\mathbf{G}\mathbf{G}'} W_{\mathbf{G}\mathbf{G}'}(\mathbf{q},\mathrm{i}\omega')\,\frac{\rho_{m\mathbf{k}-\mathbf{q}}^{n\mathbf{k}}(\mathbf{G})\rho_{m\mathbf{k}-\mathbf{q}}^{n'\mathbf{k}^*}(\mathbf{G}')}{\mathrm{i}\,(\omega+\omega')+\epsilon_{\mathrm{F}}-\epsilon_{m\mathbf{k}-\mathbf{q}}}\right) \\ &+ \delta_{nn'}\xi_{N_{\mathbf{k}}}^{\mathrm{stagger}}(\omega')\Bigg], \qquad \mathbf{k}\in\mathcal{K}. \end{aligned} \tag{8}$$

A key note is that momentum vectors $\mathbf{k}$ and $\mathbf{k}-\mathbf{q}$ are now sampled on two staggered meshes $\mathcal{K}$ and $\mathcal{K}'$, respectively, where $\mathcal{K}'$ for $\mathbf{k}-\mathbf{q}$ is the half-mesh-size shift of $\mathcal{K}$ by the choice of $\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}$.

It is worth noting that the above singularity subtraction method alone can also be applied to the standard GW calculation without using staggered meshes. We only need to simply replace $\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}$ in Eq. (8) by a Γ-centered MP mesh $\mathcal{K}_{\mathbf{q}}^{0}$, i.e.,

$$\begin{aligned} M_{\mathrm{TDL}}(\omega') &\approx \frac{|\Omega^*|}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{0}} f(\mathbf{q};\omega') + \delta_{nn'}\left(\int_{\Omega^*}\mathrm{d}\mathbf{q} - \frac{|\Omega^*|}{N_{\mathbf{k}}}\sum_{\mathbf{q}\in\mathcal{K}_{\mathbf{q}}^{0}}\right) f_{\mathrm{aux}}(\mathbf{q};\omega') \\ &= M_{N_{\mathbf{k}}}(\omega') + \delta_{nn'}\xi_{N_{\mathbf{k}}}(\omega'), \\ \mathbf{\Sigma}_{nn'}^{\mathrm{corrected}}(\mathbf{k},\mathrm{i}\omega) &= \mathbf{\Sigma}_{nn'}(\mathbf{k},\mathrm{i}\omega) - \delta_{nn'}\frac{1}{(2\pi)^4}\int_{-\infty}^{\infty}\mathrm{d}\omega'\xi_{N_{\mathbf{k}}}(\omega'). \end{aligned} \tag{9}$$

Such a correction removes the dominant $\mathcal{O}(N_{\mathbf{k}}^{-\frac{1}{3}})$ quadrature error in standard self-energy calculation, while the remaining quadrature error is $\mathcal{O}(N_{\mathbf{k}}^{-1})$ and one significant portion of it comes from calculations at zero momentum transfer $\mathbf{q}=\mathbf{0}$.

3 Numerical results

3.1 Synthetic example

Consider numerical quadrature for a non-smooth function

$$f(\mathbf{q}) = \frac{1}{|\mathbf{q}|^2} H(\mathbf{q}), \quad \mathbf{q} \in V = \left[-\frac{1}{2}, \frac{1}{2}\right]^3, \tag{10}$$

where $H(\mathbf{q})$ is smooth and compactly supported in V and $H(\mathbf{0}) \neq 0$. Such a function $H(\mathbf{q})$ is used to enforce the periodic boundary condition on ∂V and its exact definition does not matter. We test five different numerical quadrature schemes:

1. Use $\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}$ that is from the half-mesh-size shift of a Γ-centered uniform mesh so that $\mathbf{0} \notin \mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}$:

$$\frac{|\Omega^*|}{N_{\mathbf{k}}} \sum_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}} f(\mathbf{q}).$$

2. Use a uniform mesh $\mathcal{K}_{\mathbf{q}}^{0}$ that is Γ-centered, i.e., containing $\mathbf{q} = \mathbf{0}$:

$$\frac{|\Omega^*|}{N_{\mathbf{k}}} \sum_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}}^{0}}{}' f(\mathbf{q}).$$

3. Use $\mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}$ with a singularity-subtraction-based correction $\xi_{N_{\mathbf{k}}}^{\text{stagger}}$:

$$\frac{|\Omega^*|}{N_{\mathbf{k}}} \sum_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}}^{\frac{1}{2}}} f(\mathbf{q}) + \xi_{N_{\mathbf{k}}}^{\text{stagger}}.$$

4. Use $\mathcal{K}_{\mathbf{q}}^{0}$ with a singularity-subtraction-based correction $\xi_{N_{\mathbf{k}}}$:

$$\frac{|\Omega^*|}{N_{\mathbf{k}}} \sum_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}}^{0}}{}' f(\mathbf{q}) + \xi_{N_{\mathbf{k}}}.$$

5. Use $\mathcal{K}_{\mathbf{q}}^{0}$ with a head-wing-like correction which compensates the neglected integral of $f(\mathbf{q})$ in the volume element Ω_0 containing $\mathbf{q} = \mathbf{0}$:

$$\frac{|\Omega^*|}{N_{\mathbf{k}}} \sum_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}}^{0}}{}' f(\mathbf{q}) + \int_{\Omega_0} f(\mathbf{q}) \, \mathrm{d}\mathbf{q}.$$

The quadrature errors of these five schemes are plotted in Fig. 1 which illustrates the superior performance of the staggered mesh method plus a singularity-subtraction correction.

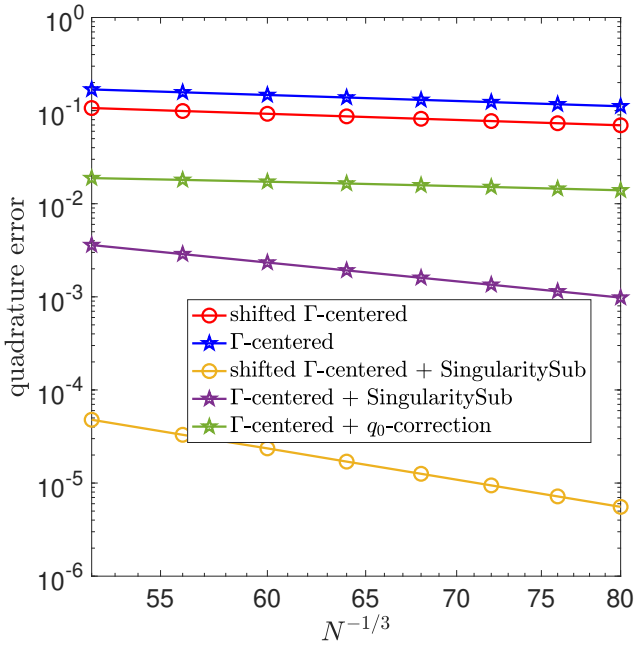


Figure 1: Absolute quadrature error of five different trapezoidal rules .

Appendix

.1 Asymptotic behavior of $W_{\mathbf{GG'}}(\mathbf{q}, \mathrm{i}\omega)$ near $\mathbf{q} = \mathbf{0}$

This section briefly summarizes Ref.[1] without using the $\mathbf{k} \cdot \mathbf{p}$ perturbation theory. For $\mathbf{G'} = \mathbf{0}$ and $\mathbf{G} \neq \mathbf{0}$, the polarizability operator's wing can be expanded at $\mathbf{q} = \mathbf{0}$ as

$$\chi_{\mathbf{G0}}(\mathbf{q}, \mathrm{i}\omega) = \mathbf{q}^T \nabla_{\mathbf{q}} \chi_{\mathbf{G0}}(\mathbf{q} = \mathbf{0}, \mathrm{i}\omega) + \mathcal{O}(|\mathbf{q}|^2).$$

For $\mathbf{G} = \mathbf{G'} = \mathbf{0}$, the polarizability operator's head can be expanded at $\mathbf{q} = \mathbf{0}$ as

$$\chi_{\mathbf{00}}(\mathbf{q}, \mathrm{i}\omega) = \frac{1}{2} \mathbf{q}^T D_{\mathbf{q}}^2 (\chi_{\mathbf{00}}(\mathbf{q} = \mathbf{0}, \mathrm{i}\omega)) \mathbf{q} + \mathcal{O}(|\mathbf{q}|^3).$$

Plugging these expansions into the dielectric operator, we have

$$\begin{aligned} \varepsilon_{\mathbf{G0}}(\mathbf{q}, \mathrm{i}\omega) &= -\frac{4\pi}{|\mathbf{G}|} \left(\frac{\mathbf{q}}{|\mathbf{q}|} \right)^T \nabla_{\mathbf{q}} \chi_{\mathbf{G0}}(\mathbf{q} = \mathbf{0}, \mathrm{i}\omega) + \mathcal{O}(|\mathbf{q}|) \\ &= \sum_{\alpha} \frac{q_\alpha}{|\mathbf{q}|} U_\alpha(\mathbf{G}, \omega) + \mathcal{O}(|\mathbf{q}|), \end{aligned} \tag{11}$$

$$\begin{aligned} \varepsilon_{\mathbf{00}}(\mathbf{q}, \mathrm{i}\omega) &= 1 - 2\pi \left(\frac{\mathbf{q}}{|\mathbf{q}|} \right)^T D_{\mathbf{q}}^2 (\chi_{\mathbf{00}}(\mathbf{q} = \mathbf{0}, \mathrm{i}\omega)) \left(\frac{\mathbf{q}}{|\mathbf{q}|} \right) + \mathcal{O}(|\mathbf{q}|) \\ &= \sum_{\alpha\beta} \frac{q_\alpha q_\beta}{|\mathbf{q}|^2} F_{\alpha\beta}(\omega) + \mathcal{O}(|\mathbf{q}|), \end{aligned} \tag{12}$$

where the two intermediate quantities are defined as

$$U(\mathbf{G}, \omega) = -\frac{4\pi}{|\mathbf{G}|} \nabla_{\mathbf{q}} \chi_{\mathbf{G0}}(\mathbf{q} = \mathbf{0}, \mathrm{i}\omega) \in \mathbb{R}^3, \tag{13}$$

$$F(\omega) = I - 2\pi D_{\mathbf{q}}^2 (\chi_{\mathbf{00}}(\mathbf{q} = \mathbf{0}, \mathrm{i}\omega)) \in \mathbb{R}^{3\times 3}. \tag{14}$$

The head and wing limits in Eqs. (11) and (12) with $\mathbf{q} \to \mathbf{0}$ are finite but depend on the direction $\hat{\mathbf{q}} = \mathbf{q}/|\mathbf{q}|$. Denote the two limits as $\mathbf{w}_{\mathbf{G}}(\hat{\mathbf{q}})$ and $H(\hat{\mathbf{q}})$. When $\mathbf{q} \to \mathbf{0}$ in direction $\hat{\mathbf{q}}$, the dielectric operator converges to

$$\varepsilon(\mathbf{q}) \to \widetilde{\varepsilon}(\hat{\mathbf{q}}) = \begin{bmatrix} H(\hat{\mathbf{q}}) & \mathbf{w}^\dagger(\hat{\mathbf{q}}) \\ \mathbf{w}(\hat{\mathbf{q}}) & \mathbf{B} \end{bmatrix},$$

where $\mathbf{B} = \varepsilon_{\mathbf{GG'}}(\mathbf{0}, \mathrm{i}\omega)$ is the body part. Using block matrix inversion, we have

$$\varepsilon^{-1}(\mathbf{q}) \to \widetilde{\varepsilon}^{-1}(\hat{\mathbf{q}}) = \begin{bmatrix} \widetilde{\varepsilon}_{\mathbf{00}}^{-1}(\hat{\mathbf{q}}) & -\widetilde{\varepsilon}_{\mathbf{00}}^{-1}(\hat{\mathbf{q}}) \mathbf{w}(\hat{\mathbf{q}})^\dagger \mathbf{B}^{-1} \\ -\mathbf{B}^{-1} \mathbf{w}(\hat{\mathbf{q}}) \widetilde{\varepsilon}_{\mathbf{00}}^{-1}(\hat{\mathbf{q}}) & \mathbf{B}^{-1} + \mathbf{B}^{-1} \mathbf{w}(\hat{\mathbf{q}}) \widetilde{\varepsilon}_{\mathbf{00}}^{-1}(\hat{\mathbf{q}}) \mathbf{w}(\hat{\mathbf{q}})^\dagger \mathbf{B}^{-1} \end{bmatrix},$$

with $\widetilde{\varepsilon}_{\mathbf{00}}^{-1}(\hat{\mathbf{q}}) = \left(H(\hat{\mathbf{q}}) - \mathbf{w}^\dagger(\hat{\mathbf{q}}) \mathbf{B}^{-1} \mathbf{w}(\hat{\mathbf{q}}) \right)^{-1}$. Define an auxiliary vector as

$$S_\alpha(\mathbf{G}) = \sum_{\mathbf{G'} \neq \mathbf{0}} \mathbf{B}_{\mathbf{GG'}}^{-1} U_\alpha(\mathbf{G'}).$$

We have

$$\widetilde{\varepsilon}^{-1}_{\mathbf{00}}(\hat{\mathbf{q}}) = \left[\sum_{\alpha\beta}\hat{\mathbf{q}}_\alpha\hat{\mathbf{q}}_\beta\left(F_{\alpha\beta} - \sum_{\mathbf{G}\neq\mathbf{0}}U^*_\alpha(\mathbf{G})S_\beta(\mathbf{G})\right)\right]^{-1} = \frac{1}{\hat{\mathbf{q}}^T\mathbf{L}_\omega\hat{\mathbf{q}}} = \frac{|\mathbf{q}|^2}{\mathbf{q}^T\mathbf{L}_\omega\mathbf{q}},$$

$$\widetilde{\varepsilon}^{-1}_{\mathbf{G0}}(\hat{\mathbf{q}}) = -\frac{\hat{\mathbf{q}}\cdot\mathbf{S}(\mathbf{G})}{\hat{\mathbf{q}}^T\mathbf{L}_\omega\hat{\mathbf{q}}} = -|\mathbf{q}|\frac{\mathbf{q}\cdot\mathbf{S}(\mathbf{G})}{\mathbf{q}^T\mathbf{L}_\omega\mathbf{q}},$$

$$\widetilde{\varepsilon}^{-1}_{\mathbf{GG'}}(\hat{\mathbf{q}}) = \mathbf{B}^{-1}_{\mathbf{GG'}} + \frac{(\hat{\mathbf{q}}\cdot\mathbf{S}(\mathbf{G}))(\hat{\mathbf{q}}\cdot\mathbf{S}^*(\mathbf{G'}))}{\hat{\mathbf{q}}^T\mathbf{L}_\omega\hat{\mathbf{q}}} = \mathbf{B}^{-1}_{\mathbf{GG'}} + \frac{(\mathbf{q}\cdot\mathbf{S}(\mathbf{G}))(\mathbf{q}\cdot\mathbf{S}^*(\mathbf{G'}))}{\mathbf{q}^T\mathbf{L}_\omega\mathbf{q}},$$

where $\mathbf{L}_\omega$ depends on ω in $\chi(\mathbf{q},\mathrm{i}\omega)$ and is symmetric positive definite (need further check). The entries of $W_{\mathbf{GG'}}(\mathbf{q},\mathrm{i}\omega)$ near $\mathbf{q}=\mathbf{0}$ then behaves as

$$W_{\mathbf{00}}(\mathbf{q},\mathrm{i}\omega) \to \frac{4\pi}{\mathbf{q}^T\mathbf{L}_\omega\mathbf{q}} - \frac{4\pi}{|\mathbf{q}|^2}, \tag{15}$$

$$W_{\mathbf{G0}}(\mathbf{q},\mathrm{i}\omega) \to -\frac{4\pi}{|\mathbf{G}|}\frac{\mathbf{q}\cdot\mathbf{S}(\mathbf{G})}{\mathbf{q}^T\mathbf{L}_\omega\mathbf{q}}, \tag{16}$$

$$W_{\mathbf{GG'}}(\mathbf{q},\mathrm{i}\omega) \to \frac{4\pi}{|\mathbf{G}||\mathbf{G'}|}\left(\mathbf{B}^{-1}_{\mathbf{GG'}} + \frac{(\mathbf{q}\cdot\mathbf{S}(\mathbf{G}))(\mathbf{q}\cdot\mathbf{S}^*(\mathbf{G'}))}{\mathbf{q}^T\mathbf{L}_\omega\mathbf{q}} - \delta_{\mathbf{GG'}}\right). \tag{17}$$

where $\mathbf{L}_\omega, \mathbf{S}, \mathbf{B}$ all depend on ω.

.2 Gradient and Hessian of the polarizability operator

The singularity-subtraction-based corrections for both standard and staggered GW calculations Eqs. (8) and (9)require the key quantity $\mathbf{L}_\omega$ defined as

$$\mathbf{L}_\omega(\alpha,\beta) = F_{\alpha\beta} - \sum_{\mathbf{G},\mathbf{G'}\neq\mathbf{0}} U^*_\alpha(\mathbf{G})\mathbf{B}^{-1}_{\mathbf{GG'}}U_\beta(\mathbf{G'}),$$

where the two intermediate quantities $U_\alpha(\mathbf{G})$ and $F_{\alpha\beta}$ in Eqs. (13) and (14) further depend on the gradient and Hessian of the polarizaibility operator $\chi_{\mathbf{GG'}}(\mathbf{q},\mathrm{i}\omega)$ at $\mathbf{q}=\mathbf{0}$.

.2.1 $\mathbf{k}\cdot\mathbf{p}$ perturbation theory

Using $\mathbf{k}\cdot\mathbf{p}$ perturbation theory[3], we have

$$\rho^{i\mathbf{k}}_{a\mathbf{k}-\mathbf{q}}(\mathbf{0})\big|_{\mathbf{q}\to\mathbf{0}} = \mathrm{i}\mathbf{q}\cdot\langle\psi_{i\mathbf{k}}|\mathbf{r}|\psi_{a\mathbf{k}}\rangle = \frac{\mathrm{i}\mathbf{q}\cdot\langle\psi_{i\mathbf{k}}|\nabla|\psi_{a\mathbf{k}}\rangle}{\epsilon_{a\mathbf{k}}-\epsilon_{i\mathbf{k}}}, \tag{18}$$

which leads to the computation of $U_\alpha(\mathbf{G})$ and $F_{\alpha\beta}$ as

$$F_{\alpha\beta}(\omega) = \delta_{\alpha\beta} - \frac{8\pi}{N_\mathbf{k}\Omega}\sum_{\mathbf{k}}\sum_{i}^{\mathrm{occ}}\sum_{a}^{\mathrm{vir}}\frac{\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}}{\omega^2+(\epsilon_{i\mathrm{k}}-\epsilon_{a\mathbf{k}})^2}\langle\psi_{i\mathbf{k}}|r_\alpha|\psi_{a\mathbf{k}}\rangle\langle\psi_{a\mathbf{k}}|r_\beta|\psi_{i\mathbf{k}}\rangle,$$

$$U_\alpha(\mathbf{G},\omega) = \frac{4\pi}{|\mathbf{G}|}\frac{2i}{N_\mathbf{k}\Omega}\sum_{\mathbf{k}}\sum_{i}^{\mathrm{occ}}\sum_{a}^{\mathrm{vir}}\frac{\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}}{\omega^2+(\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}})^2}\rho^{i\mathbf{k}}_{a\mathbf{k}}(\mathbf{G})\langle\psi_{a\mathbf{k}}|r_\alpha|\psi_{i\mathbf{k}}\rangle.$$

However, this theory requires the complete basis limit (or at least the periodic basis is independent of $\mathbf{k}$) which is not the case when using the crystalline Gaussian basis functions.

.2.2 Perturbation theory from Lin's note

The following definition of pair product differs from that in Tianyu's paper by a minus sign:

$$\rho_{n\mathbf{k},n'\mathbf{k}'}(\mathbf{G}) = \langle \psi_{n\mathbf{k}} | e^{\mathrm{i}(\mathbf{k}-\mathbf{k}'-\mathbf{G})\mathbf{r}} | \psi_{n'\mathbf{k}'} \rangle = \langle u_{n\mathbf{k}} | e^{-\mathrm{i}\mathbf{G}\mathbf{r}} | u_{n'\mathbf{k}'} \rangle ,$$

where $u_{n\mathbf{k}} = e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\psi_{n\mathbf{k}}$ denotes the periodic part of crystalline orbital $\psi_{n\mathbf{k}}$. Note that

$$\sum_{ia} \rho_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{G}) \rho^*_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{G}') = \sum_{ia} \left\langle u_{i\mathbf{k}} | e^{-\mathrm{i}\mathbf{G}\cdot\mathbf{r}} | u_{a(\mathbf{k}+\mathbf{q})} \right\rangle \left\langle u_{a(\mathbf{k}+\mathbf{q})} | e^{\mathrm{i}\mathbf{G}'\cdot\mathbf{r}} | u_{i\mathbf{k}} \right\rangle ,$$

which is gauge-invariant for any $\mathbf{G}$ and $\mathbf{G}'$ and

$$\nabla_{\mathbf{q}} \left(\rho_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{G}) \rho^*_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{G}') \right) |_{\mathbf{q}=\mathbf{0}} = \langle u_{i\mathbf{k}} | \, e^{-\mathrm{i}\mathbf{G}\cdot\mathbf{r}} \left\{ \nabla_{\mathbf{q}} \left(|u_{a(\mathbf{k}+\mathbf{q})}\rangle \langle u_{a(\mathbf{k}+\mathbf{q})}| \right) |_{\mathbf{q}=\mathbf{0}} \right\} e^{\mathrm{i}\mathbf{G}'\cdot\mathbf{r}} \, |u_{i\mathbf{k}}\rangle .$$

When $\mathbf{G}' = \mathbf{0}$, using perturbation theory (see Lin's note for details) can give

$$\partial_{\mathbf{q}_\alpha} \left(\rho_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{G}) \rho^*_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{0}) \right) |_{\mathbf{q}=\mathbf{0}} = \rho_{i\mathbf{k},a\mathbf{k}}(\mathbf{G}) C^*_{\mu a}(\mathbf{k}) \left(\frac{\partial_{\mathbf{k}_\alpha} H(\mathbf{k}) - \epsilon_{a\mathbf{k}} \partial_{\mathbf{k}_\alpha} S(\mathbf{k})}{\epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}}} + T_\alpha(\mathbf{k}) \right)_{\mu\nu} C_{\nu i}(\mathbf{k})$$

where μ, ν indicates atomic Gaussian orbitals and the crystalline Gaussian orbitals are defined as

$$\phi_{\mu\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}\in\mathbb{L}} e^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}} \mu(\mathbf{r}-\mathbf{R}).$$

The three involved operators above are defined as

$$H_{\mu\nu}(\mathbf{k}) = \langle \phi_{\mu\mathbf{k}} | -\frac{1}{2}\Delta + V_{\text{eff}} | \phi_{\nu\mathbf{k}} \rangle , \tag{19}$$

$$S_{\mu\nu}(\mathbf{k}) = \langle \phi_{\mu\mathbf{k}} | \phi_{\nu\mathbf{k}} \rangle , \tag{20}$$

$$T_{\mu\nu}(\mathbf{k}) = \sum_{\mathbf{R}\in\mathbb{L}} e^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}} \int \mu(\mathbf{r})(\mathrm{i}\mathbf{r})\nu(\mathbf{r}-\mathbf{R}) \, \mathrm{d}\mathbf{r}, \tag{21}$$

and $C_{\mu n}(\mathbf{k})$ is the molecular coefficient that solves the generalized eigenvalue problem:

$$\sum_{\nu} H_{\mu\nu}(\mathbf{k}) C_{\nu n}(\mathbf{k}) = \epsilon_{n\mathbf{k}} \sum_{\nu} S_{\mu\nu}(\mathbf{k}) C_{\nu n}(\mathbf{k}).$$

The second order derivative of the polarizability operator can also be derived based on perturbation theory (see Lin's note) as,

$$\begin{aligned}
&\partial^2_{q_\alpha q_\beta} \left(\rho_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{0}) \rho^*_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{0}) \right) \Big|_{\mathbf{q}=\mathbf{0}} \\
=&2 \, \mathrm{Re} \left(C^*_{\mu i}(\mathbf{k}) \left(T^*_\alpha(\mathbf{k}) + \frac{\partial_{\mathbf{k}_\alpha} H(\mathbf{k}) - \epsilon_{a\mathbf{k}} \partial_{\mathbf{k}_\alpha} S(\mathbf{k})}{\epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}}} \right)_{\mu\nu} C_{\nu a}(\mathbf{k}) \right) \\
&\times \left(C^*_{\mu a}(\mathbf{k}) \left(T_\beta(\mathbf{k}) + \frac{\partial_{\mathbf{k}_\beta} H(\mathbf{k}) - \epsilon_{a\mathbf{k}} \partial_{\mathbf{k}_\beta} S(\mathbf{k})}{\epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}}} \right)_{\mu\nu} C_{\nu i}(\mathbf{k}) \right)
\end{aligned}$$

When the periodic basis functions $e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\phi_{\mu\mathbf{k}}$ are independent of $\mathbf{k}$, we have $T(\mathbf{k}) = \nabla_{\mathbf{k}} S(\mathbf{k}) = \mathbf{0}$, and the above perturbation result for gradients is reduced to

$$\begin{aligned}\nabla_{\mathbf{q}} \left(\rho_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{G}) \rho^*_{i\mathbf{k},a(\mathbf{k}+\mathbf{q})}(\mathbf{0}) \right) |_{\mathbf{q}=\mathbf{0}} &= \rho_{i\mathbf{k},a\mathbf{k}}(\mathbf{G}) \left(\frac{C^*_{\mu a}(\mathbf{k}) \nabla_{\mathbf{k}} H_{\mu\nu}(\mathbf{k}) C_{\nu i}(\mathbf{k})}{\epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}}} \right) \\ &= \rho_{i\mathbf{k},a\mathbf{k}}(\mathbf{G}) \left(\frac{-\mathrm{i} \langle \psi_{a\mathbf{k}} | \nabla | \psi_{i\mathbf{k}} \rangle}{\epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}}} \right)\end{aligned}$$

which is the same as the $\mathbf{k} \cdot \mathbf{p}$ perturbation theory.

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